Week11_group_and_individual_assignment_Sulaiman (Stan) Abu Romman

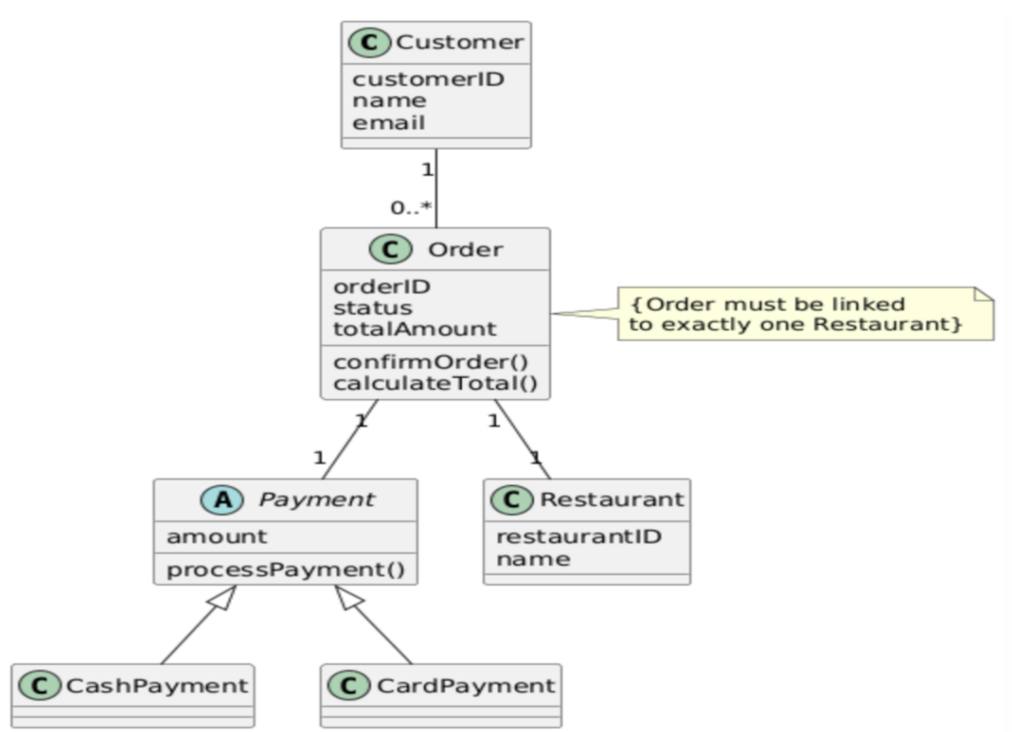
Part 1 – Group Assignment: Fine Tune Your Design Class Diagram

In this assignment, the Design Class Diagram of the Stomach Saver system is reviewed and refined by applying advanced UML concepts. Fine-tuning the DCD improves clarity, correctness, and prepares the design for implementation. The refinements include the use of constraints, stereotypes, inheritance, and polymorphism.

Constraints are used to express logical rules that must always be respected by the system. In the Stomach Saver system, constraints ensure valid relationships between orders, payments, and restaurants. Stereotypes are applied to clarify the role of classes and relationships and to extend standard UML notation.

Inheritance and polymorphism are used where shared behavior exists. This supports the Open–Closed Principle by allowing the system to be extended without modifying existing classes. The refined DCD reflects a more flexible and maintainable design.

Refined Design Class Diagram – Stomach Saver



Part 2 – Individual Assignment: Reflection on Fine-Tuning the DCD

Fine-tuning the Design Class Diagram improves the overall quality of the system design. By introducing abstract classes, inheritance, and constraints, the diagram becomes more expressive and closer to a real implementation. The use of an abstract Payment class allows different payment methods to be added without changing the Order class, which supports polymorphism and design for change.

Constraints help document business rules directly in the model, reducing ambiguity. Stereotypes and refined relationships make the diagram easier to understand and maintain. These improvements ensure that the DCD accurately represents system behavior and is consistent with object-oriented design principles such as low coupling, high cohesion, and the Open–Closed Principle.